

What is a Computer ?

A computer is an electronic device that manipulates information, or data. It has the ability to store, retrieve, and process data. You may already know that you can use a computer to type documents, send email, play games, and browse the Web.



What is a Network ?

A network consists of two or more computers that are linked in order to share resources (such as printers and CDs), exchange files, or allow electronic communications. The computers on a network may be linked through cables, telephone lines, radio waves, satellites, or infrared light beams.

Two very common types of networks include:

- [Local Area Network \(LAN\)](#)
- [Wide Area Network \(WAN\)](#)

Networks : What is a network?

A Network is formed when 2 or more computers are linked together.

Every computer or device on a network can send and receive data from any of the other computers or devices connected to the network.



A computer which is not part of a network is called a **stand-alone computer**.



What is a LAN ?

A local area network (LAN) is a collection of devices connected together in one physical location, such as a building, office, or home. A LAN can be small or large, ranging from a home network with one user to an enterprise network with thousands of users and devices in an office or school.

LAN connects devices that are in a single, limited area.

What is a WAN ?

A WAN is a collection of [local-area networks](#) (LANs) or other networks that communicate with one another. A WAN is essentially a network of networks, with the Internet the world's largest WAN.

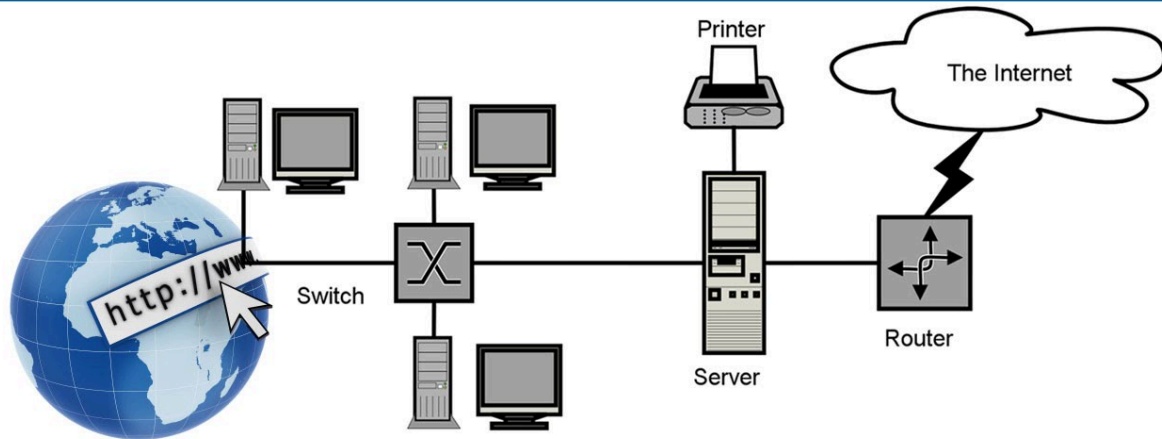


What is an Internet ?

The full form of the internet is **an interconnected network (Network of Networks)**.
The Internet is **a vast network that connects computers all over the world**.

The Internet is **a global network of billions of computers and other electronic devices**. With the Internet, it's possible to access almost any information, communicate with anyone else in the world, and do much more. You can do all of this by connecting a computer to the Internet, which is also called going online.

THE INTERNET IS...



What is an Application?

An application (more commonly known as **an app**) , also referred to as an application program or application software, is **a computer software package** . An application can be self-contained or a group of programs.

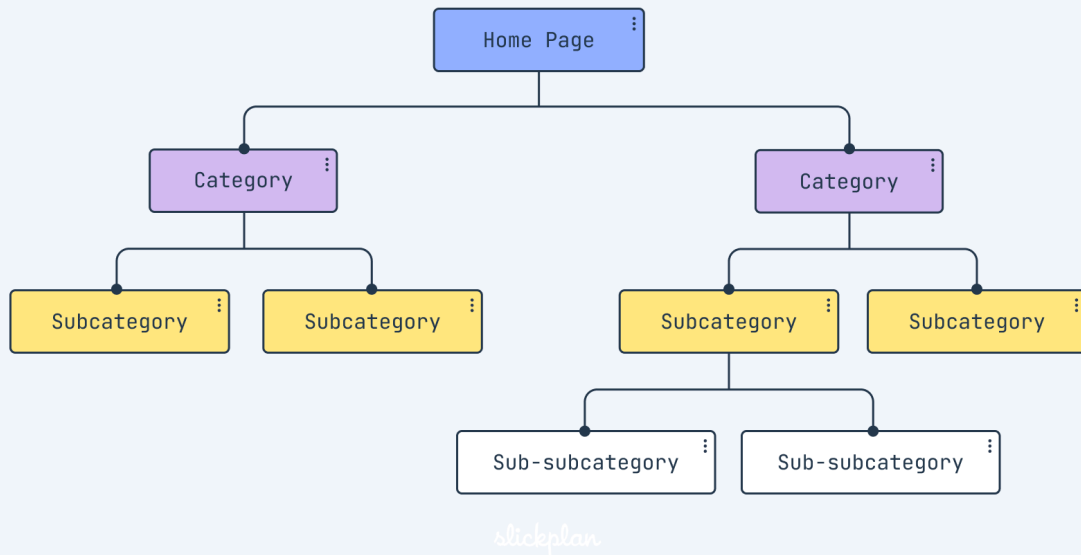


What is a Web Application?

A web application (web app) is an application program that is stored on a remote server and delivered over the internet through a browser interface.

A web application is software that runs in your web browser. Businesses have to exchange information and deliver services remotely. They use web applications to connect with customers conveniently and securely.

Website structure



Website

It has static content, with the main focus of informing the user about the business vision/products/benefits.

Only Read Mode, the user is unable to manipulate the information.

The website does not need to be pre-compiled before deployment.

The website's function and purpose are quite simple.

The website doesn't require user authentication.



Web Application

It has dynamic content, designed specifically for interaction with the end-users.

Read and Write Mode, the user is able to read and manipulate the information.

This web app should be pre-compiled before the deployment.

The web application's function and purpose are rather complex.

Most web applications require user authentication.

What is a Web Browser ?

A software application used to access information on the World Wide Web is called a Web Browser. When a user requests some information, the web browser fetches the data from a web server and then displays the webpage on the user's screen.

A web browser **takes you anywhere on the internet, letting you see text, images and video** from anywhere in the world.



Safari

Apple

MacOS, iOS



Firefox

Mozilla

MacOS, MS Windows, Linux OS,
Android OS



Chrome

Google

MacOS, MS Windows, Linux OS,
Android OS, Chrome OS



Edge new

Microsoft

MS Windows, MacOS, iOS,
Android OS



Opera

Opera Software

MacOS, MS Windows, Linux OS,
Android OS



What is a Server ?

A server is a computer or system that provides resources, data, services, or programs to other computers, known as clients, over a network.

Types of Servers



Web server



Mail server



Application server



Database server



DNS server



Proxy server



DHCP server



File server



Gaming server



Print server

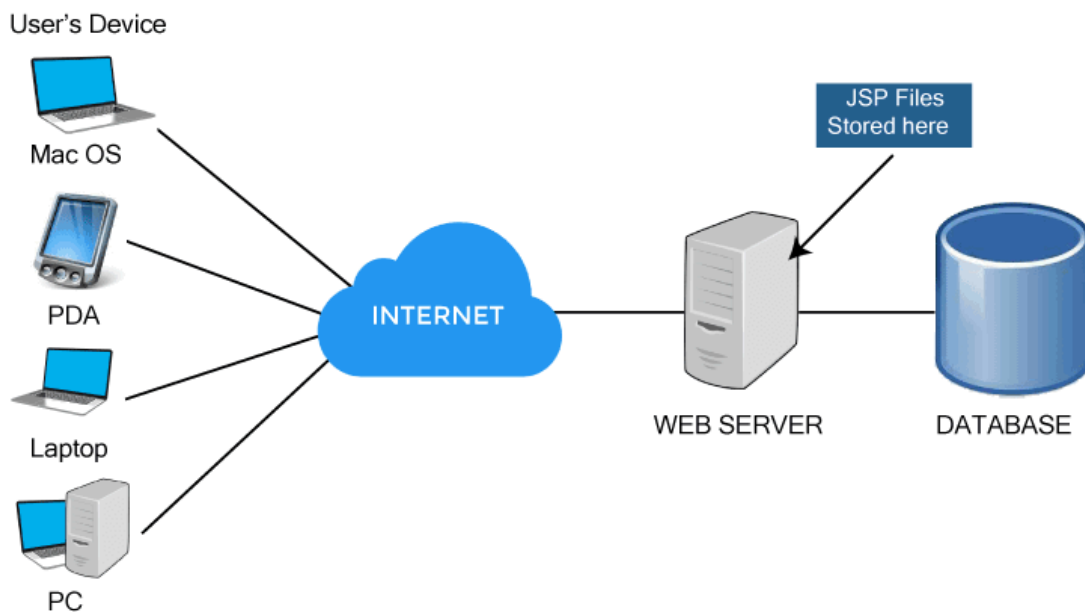


What is a Client v/s Customer ?

A customer is someone who buys products or services from a company, while a client refers to a certain type of customer who purchases professional services from a business.

What is a Web Server ?

A web server is software and hardware that uses HTTP (Hypertext Transfer Protocol) and other protocols to respond to client requests made over the World Wide Web. The main job of a web server is to display website content through storing, processing and delivering webpages to users.

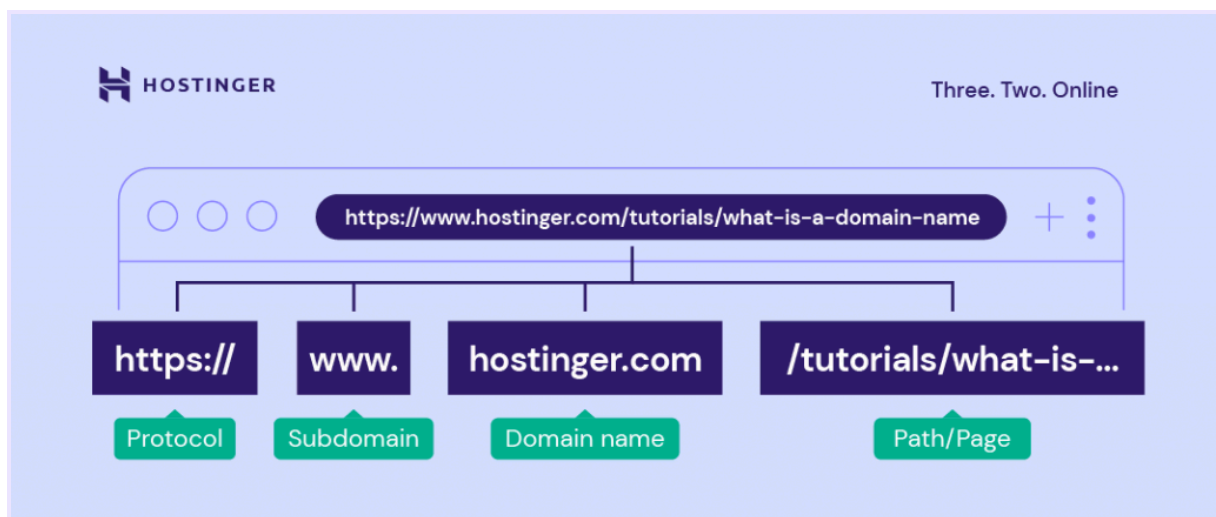


What is HTTP ?

The Hypertext Transfer Protocol (HTTP) is the foundation of the World Wide Web, and is used to load webpages using hypertext links.

What is a URL ?

A URL, short form for a uniform resource locator is a web address pointing to a specific website, a web page, or a document on the internet.



A URL, also known as a web address, identifies a web resource as well as its location on a computer network and how to access it (e.g. `http://www.library.pitt.edu`). A URN is a formal naming scheme that identifies a resource, but does not indicate its location or how to access it.

What is a Web Framework ?

A **web framework (WF)** or **web application framework (WAF)** is a [software framework](#) that is designed to support the development of [web applications](#) including web services, [web](#)

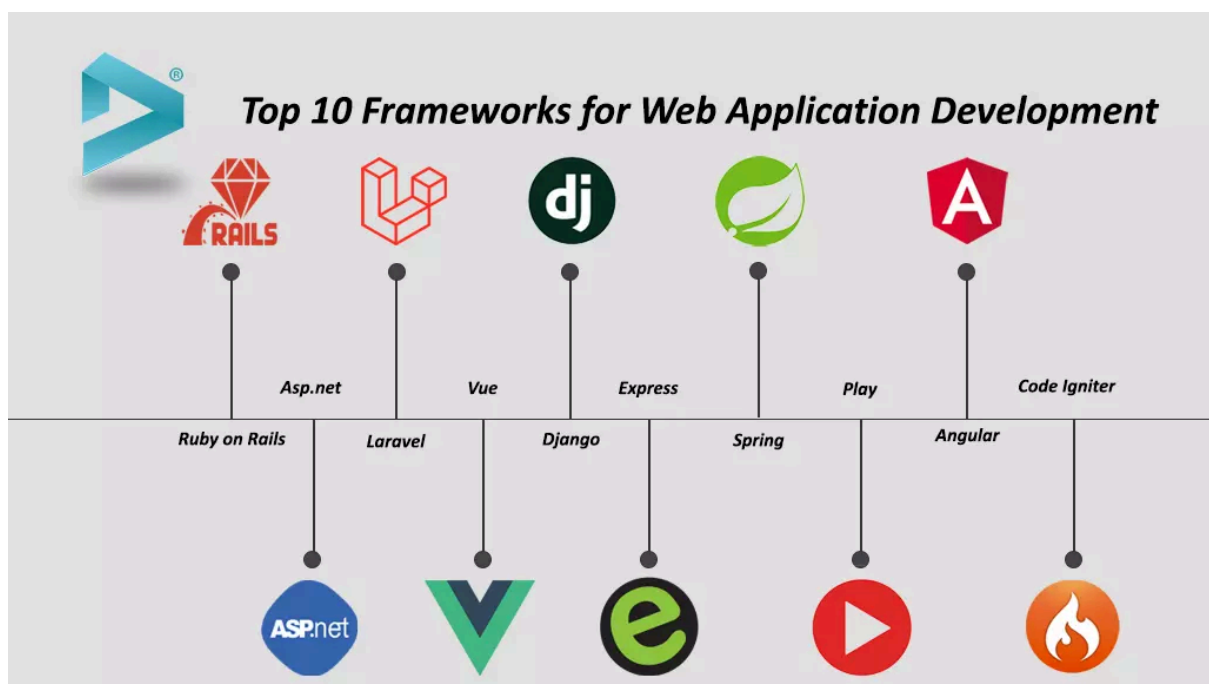
[resources](#), and [web APIs](#). Web frameworks provide a standard way to build and deploy web applications on the [World Wide Web](#)



What is a
Web Application Framework?

A software framework for building web applications efficiently.

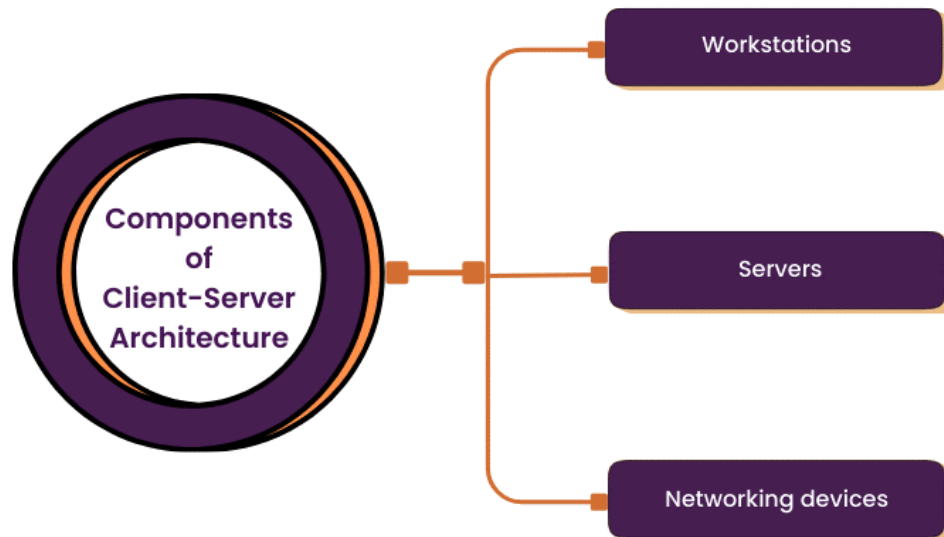
SPACE
TECHNOLOGIES



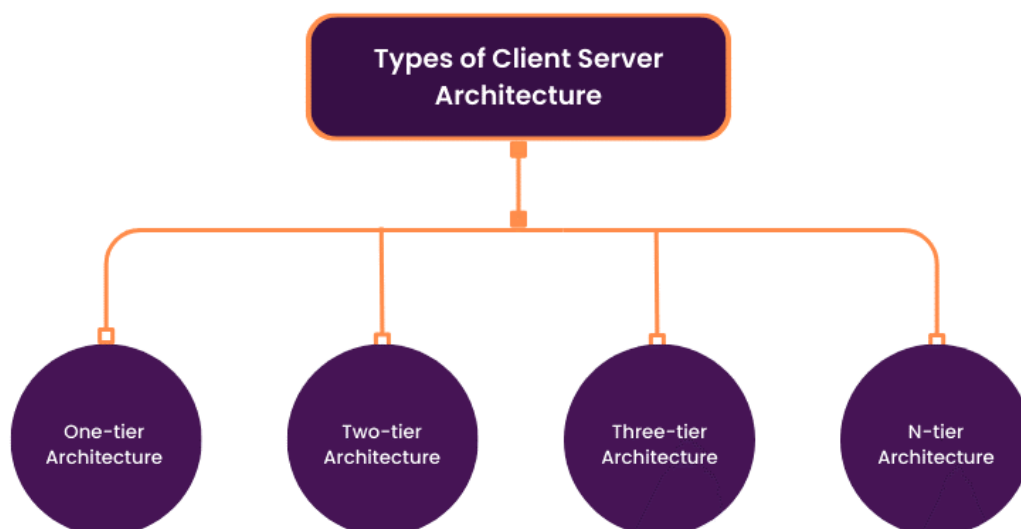
What is a Client-Server Architecture ?

Client-server architecture is a type of computer network where multiple clients request and receive files and services from a centralized server(host computer) over a local or internet connection. A client uses an application as an interface to connect to the server.

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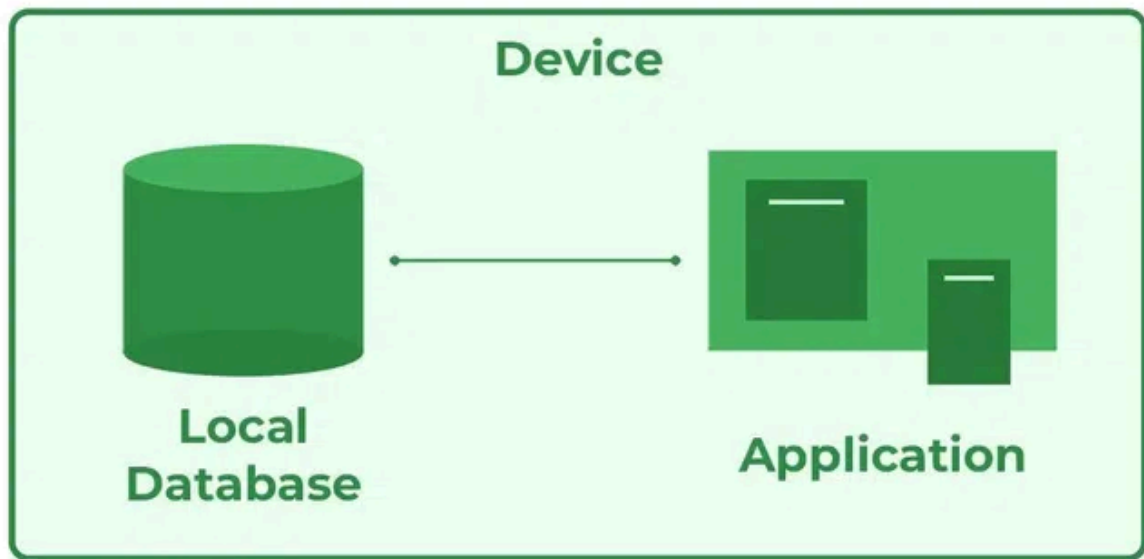


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1-Tier Architecture

In 1-Tier Architecture the database is directly available to the user, the user can directly sit on the DBMS and use it that is, the client, server, and Database are all present on the same machine.

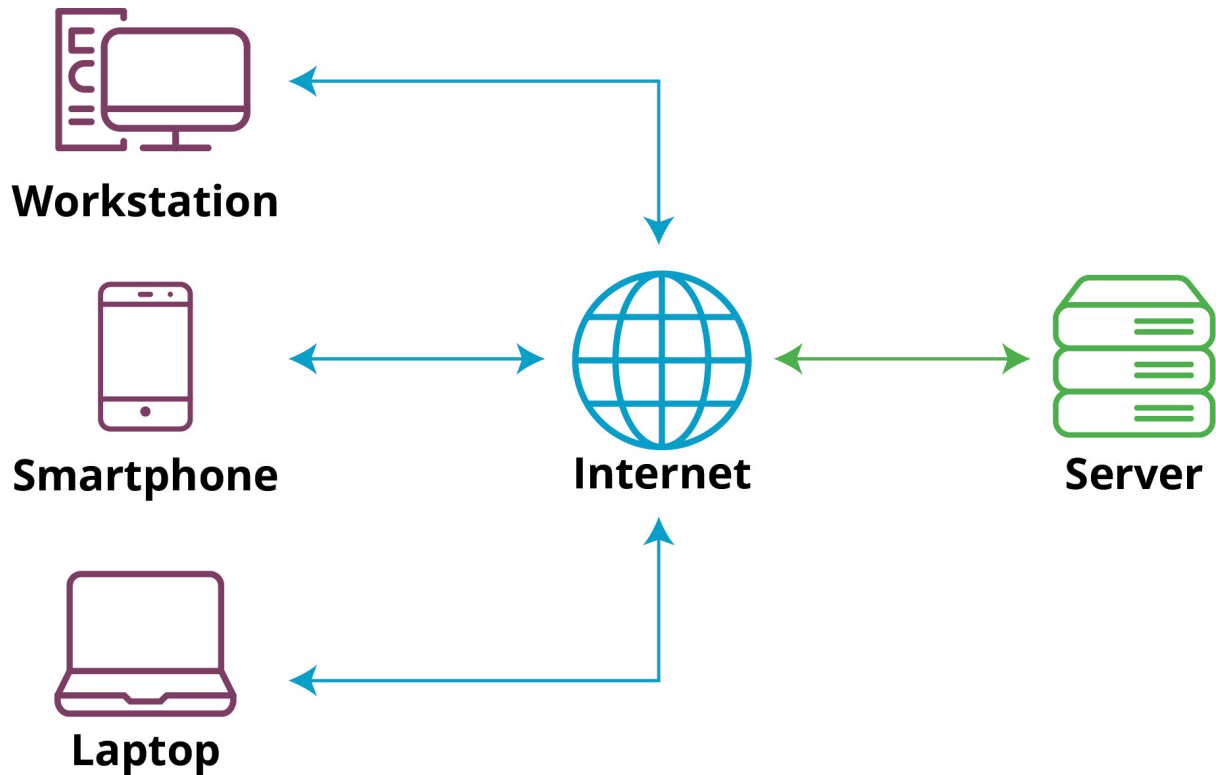


Advantages : Simple Architecture, Cost-Effective & Easy to implement.

Examples : Offline Desktop Apps, Offline Mobile Apps

2-Tier Architecture

The 2-tier architecture is similar to a basic [client-server model](#). The application at the client end directly communicates with the database on the server side. APIs like ODBC and JDBC are used for this interaction. The server side is responsible for providing query processing and transaction management functionalities. On the client side, the user interfaces and application programs are run. The application on the client side establishes a connection with the server side to communicate with the DBMS.

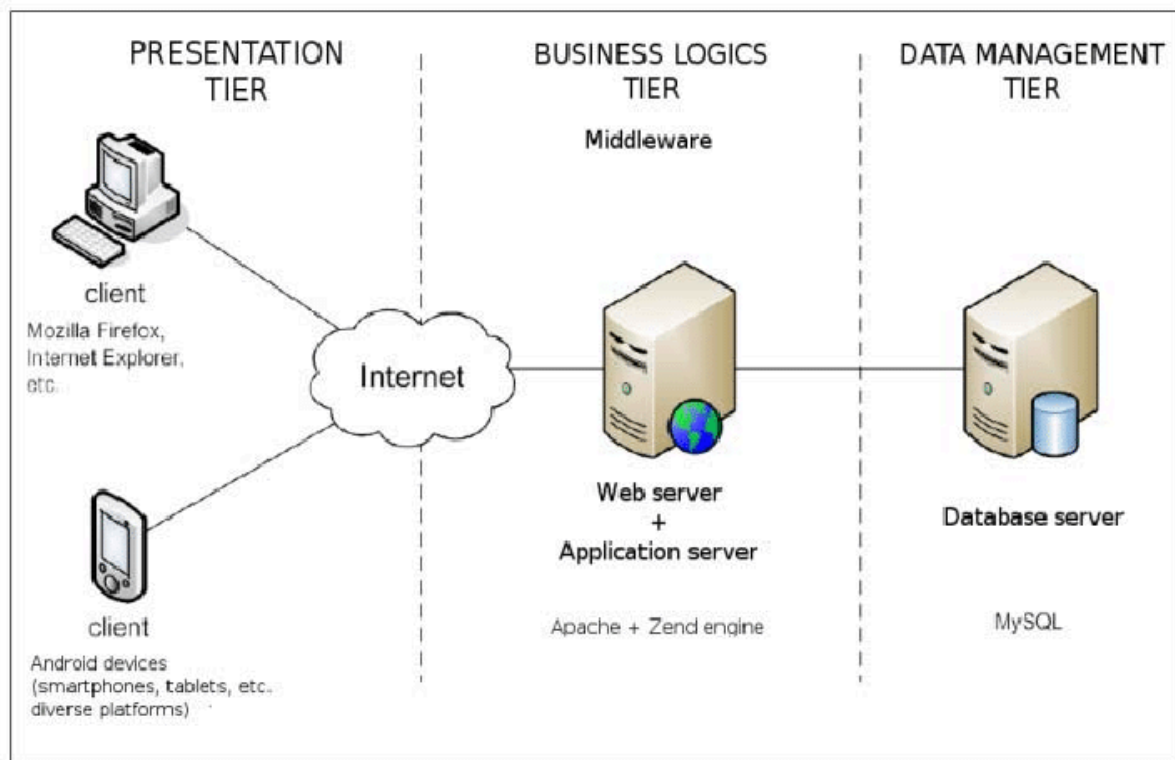


Advantages : Easy to access, Scalable, Low Cost, Easy Deployment & Simple

Example: Banking Application, Email (Outlook), Reservation Systems,

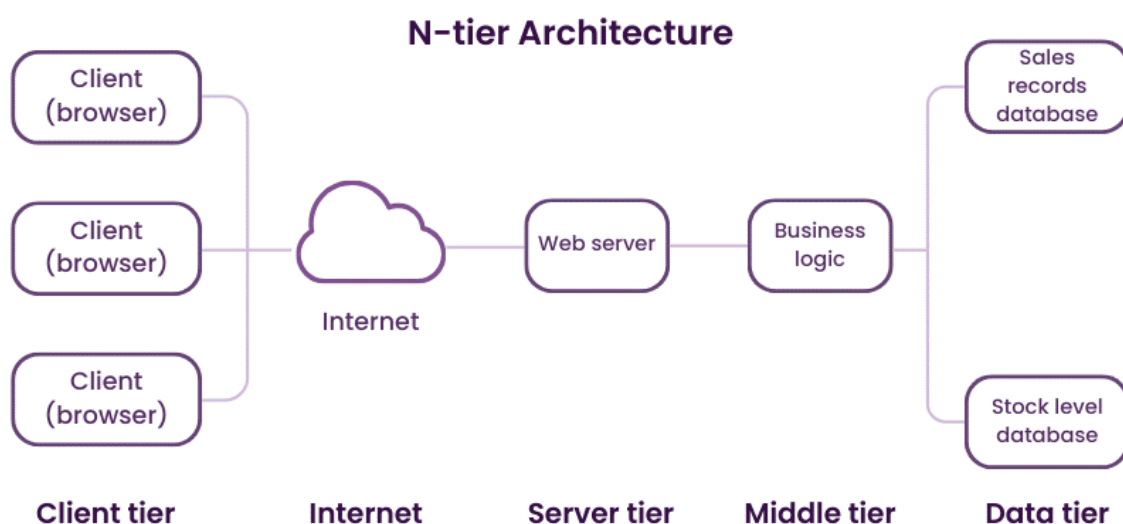
3-Tier Architecture

In [3-Tier Architecture](#), there is another layer between the client and the server. The client does not directly communicate with the server. Instead, it interacts with an application server which further communicates with the database system and then the query processing and transaction management takes place. This intermediate layer acts as a medium for the exchange of partially processed data between the server and the client. This type of architecture is used in the case of large web applications.



Advantages : Enhanced Scalability, Data Integrity & Security

Examples : Amazon, Netflix, Facebook, Google Search, Twitter, LinkedIn, Uber, Airbnb, Salesforce, Github

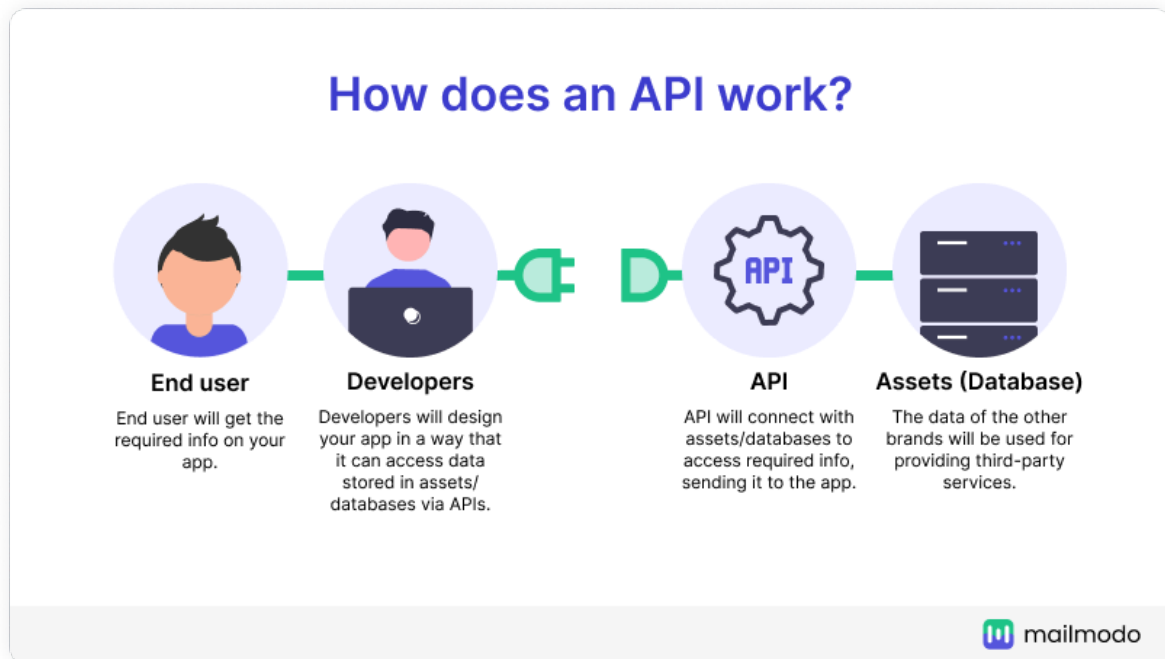


What is a API ?

API stands for **Application Programming Interface**.

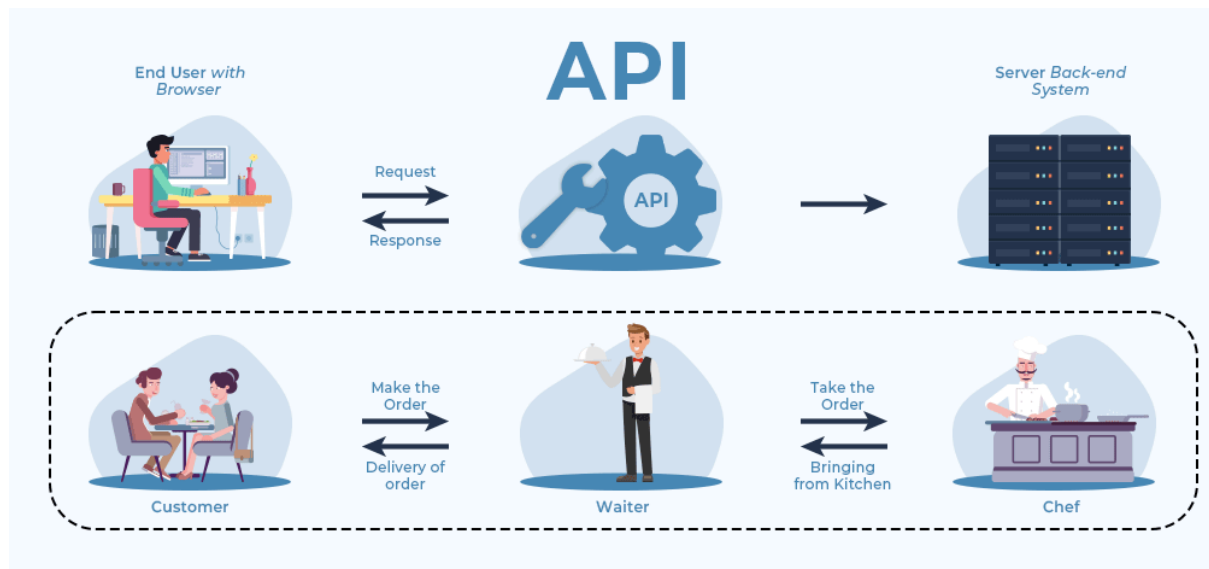
An API, or application programming interface, is a set of rules or protocols that let software applications communicate with each other to exchange data, features and functionality.

- a software intermediary that allows two applications to talk to each other. APIs are an accessible way to extract and share data within and across organizations.



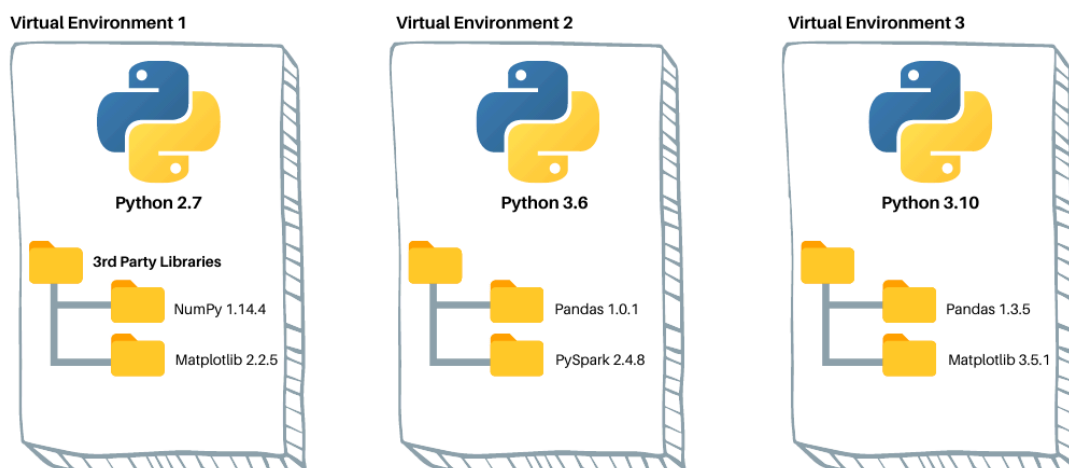
Is server and API same?

The API validates the request and passes the request to the destination server or program. The server sends a response back, first through the API, which then transfers it back to the client.



What is a Virtual Environment ?

A Python Virtual Environment is an isolated space where you can work on your Python projects, separately from your system-installed Python. You can set up your own libraries and dependencies without affecting the system Python.

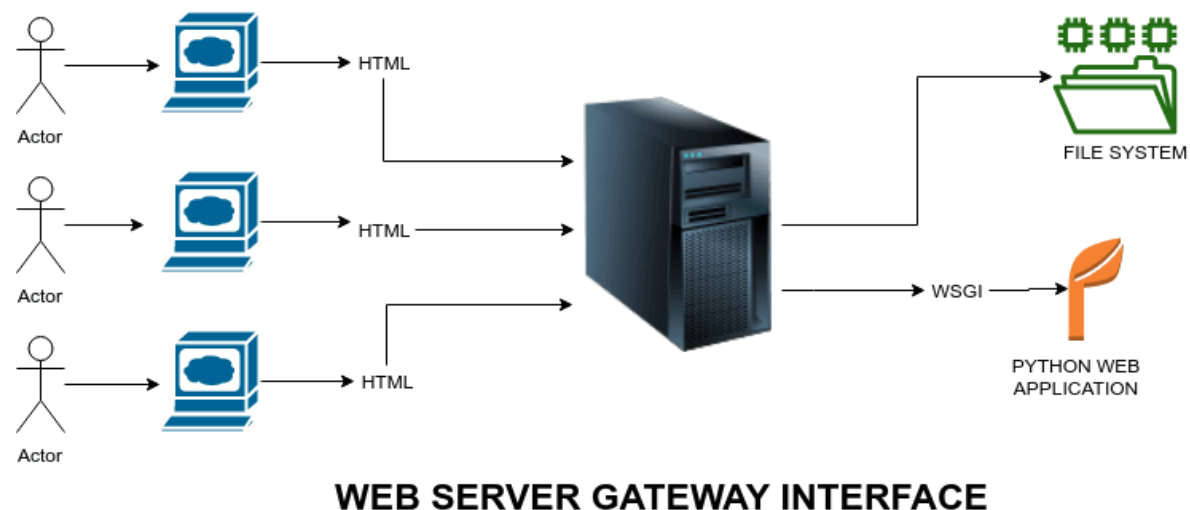


Gateway: Serving as the entry point for API requests, responsible for tasks like routing, load balancing, and request and response transformation.

WSGI and ASGI are Python specifications that define how web servers can communicate with web applications written in Python.

What is a WSGI ?

Web Server Gateway Interface (WSGI) is a mediator responsible for conveying communication between a web server and a Python web application. It explains how the web server communicates with the app and how the app can be chained for processing a request.



What is a ASGI ?

ASGI (Asynchronous Server Gateway Interface) is a spiritual successor to WSGI, intended to provide a standard interface between async-capable Python web servers, frameworks, and applications.

WSGI	ASGI
Synchronous	Asynchronous
No (Sequential Processing)	Yes (Supports Asynchronous Operations)
Traditional Web Apps	Real-time Apps, WebSockets, Streaming, etc.
Request-Response Cycle	Supports Various Protocols (incl. Async)
Synchronous Middleware	Asynchronous Middleware
Widely Adopted, Mature	Emerging, Increasing Adoption

What is a MVC ?

The [Model-View-Controller \(MVC\)](#) framework is an architectural/design pattern that separates an application into three main logical components **Model**, **View**, and **Controller**. Each architectural component is built to handle specific development aspects of an application. It isolates the business logic and presentation layer from each other. It was traditionally used for desktop graphical user interfaces (GUIs).

